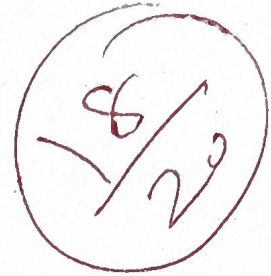


SIMATS ENGINEERING
ASSESSMENT TOOL 3
MCQ Test



COURSE NAME: Computer Networks

COURSE CODE: CSA07/4

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 20%

QUESTIONS:20

Total Marks:20

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?
a) Slow Start
b) Fast Retransmit
☒ c) Fast Recovery
d) Tahoe Reset
(BL4)(1)
2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:
☒ a) 9 MSS
b) 16 MSS
c) 12 MSS
d) 8 MSS
(BL4)(1)
3. Which field in the TCP header is used to implement flow control at the receiver side?
a) Sequence number
b) Acknowledgement number
☒ c) Window size
d) Urgent pointer
(BL4)(1)
4. Nagle's algorithm causes increased latency primarily because it:
a) Disables ACK piggybacking
☒ b) Buffers small segments until ACK arrives
c) Reduces CWND on every packet

~~with high RTT backlog~~

~~(BL4)(1)~~

6. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

~~a) MTU size~~

~~b) cwnd growth rate relative to RTT~~

~~c) DNS resolution time~~

~~d) UDP competition~~

~~(BL4)(1)~~

7. UDP is preferred over TCP for DNS queries primarily because:

~~a) UDP supports fragmentation~~

~~b) DNS responses are small and connection overhead is unnecessary~~

~~c) TCP cannot use port 53~~

~~d) UDP has better error correction~~

~~(BL4)(1)~~

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:

~~a) Increase UDP buffer size~~

~~b) Switch to TCP~~

~~c) Implement Forward Error Correction (FEC)~~

~~d) Reduce frame rate to 1 fps~~

~~(BL4)(1)~~

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?

~~a) 1480 bytes~~

~~b) 1472 bytes~~

~~c) 1460 bytes~~

~~d) 1500 bytes~~

~~(BL4)(1)~~

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:

~~a) Eliminating IP header overhead~~

~~b) 0-RTT or 1-RTT connection setup~~

~~c) Removing encryption~~

~~d) Native multicast support~~

~~(BL4)(1)~~

10. HTTP/1.1 persistent connections reduce latency by:

~~a) Enabling UDP fallback~~

~~b) Reusing the same TCP connection for multiple requests~~

~~c) Compressing HTTP headers~~

~~d) Sending requests without waiting for DNS~~

~~(BL4)(1)~~

11. Head-of-line blocking in HTTP/1.1 occurs when:

~~a) The DNS server is slow~~

- (c) A response blocks subsequent responses on the same connection
(d) TCP RWND drops to zero
(e) The server rejects pipelined requests
(BL4)(1)
12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:
- a) Application layer
 - b) DNS layer
 - c) TCP layer
 - d) IP layer
- (BL4)(1)
13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?
- a) 301
 - b) 302
 - c) 304
 - d) 307
- (BL4)(1)
14. In HTTP caching, the Cache-Control: no-cache directive means:
- a) Never cache this resource
 - b) Validate with the server before using a cached copy
 - c) Cache for 0 seconds
 - d) Disable all caching headers
- (BL4)(1)
15. Which DNS record type maps a domain name to an IPv6 address?
- a) A
 - b) CNAME
 - c) AAAA
 - d) MX
- (BL4)(1)
16. A DNS resolver switches from UDP to TCP when:
- a) The query type is MX
 - b) The response is truncated (TC bit set)
 - c) TTL exceeds 3600 s
 - d) The authoritative server is unreachable
- (BL4)(1)
17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?
- a) ARP poisoning
 - b) DNS cache poisoning
 - c) SYN flooding
 - d) BGP hijacking
- (BL4)(1)
18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:
- a) Cache hit ratio

RUBRICS FOR CALCULATION

Score	Weightage %	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Conceptual Understanding	30%	Demonstrates thorough understanding & answers all questions accurately	Shows good understanding with few errors	Partial understanding & several incorrect responses	Lacks understanding & majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions